CYBERSECURITY AWARENESS WORKSHEET

Spot the Secure Website: HTTP vs HTTPS

Worksheet completed by Charlotte

This activity helps participants identify secure vs insecure websites by analysing URL structure, browser indicators, and trust signals.

The goal is to improve awareness of how to recognise potentially unsafe websites and understand that HTTPS alone does not guarantee legitimacy.

Section 1: Website Security Evaluation

Website	URL	Indicators	Secure? (Yes/No)
A	http://online-banking-login.example.com	No padlock, HTTP	No
B	https://university.edu/login	Padlock, HTTPS	Yes
C	https://secure-paypal-login.example.net	HTTPS but suspicious domain	No

Questions:

Which website is most secure? Why?

B, as it uses https and has a padlock enabled.

Why is HTTP considered unsafe?

Does encrypt messages.

Does HTTPS always mean a site is trustworthy? Explain.

No – Doesn't ensure the website is actually safe.

Section 2: Safe vs Unsafe Links

Evaluate the following links-

Task:

- Mark each link as Safe / Suspicious
- Identify one red flag for each suspicious link

1. https://paypal.com/login

Safe – looks normal

2. https://paypal-secure-login.verify-now.example.com

Suspicious – Long URL

3. https://uwa.edu.au

Safe – looks normal

4. http://free-prizes.example.net

Insecure – uses http and offers free prizes

Section 3: File Safety (Malware Awareness)

Determine whether these files are safe:

- invoice.pdf.exe
- report.docx
- update_patch_v2.zip
- photo.png

Questions:

- Which file is most suspicious? Why?
- What is a common trick used in malicious file names?

Zip file, that could contain potential malware.

Tricks – Dodgy name.

Section 4: Password Strength Analysis

Rank the following passwords from weakest → strongest:

☐1 123456

☐3 password2024

☐2 sam12345

☐4 UWA!secure9

☐5 Xy9#kL2pQ

Questions:

- What makes a password strong?
- Why is password reuse dangerous?

Strong – Length, variety

Reuse – Same password can access different accounts if discovered

Section 5: Social Engineering Scenarios

Scenario 1:

You receive a message from “IT Support” asking for your password to fix an issue.

Scenario 2:

A friend messages you urgently asking for money.

Questions:

- What are the warning signs?
- What would you do in each situation?

1 – Dodgy asking for passwords, and I would block them

2 – Dodgy asking for money, and I would ask them in person

Section 6: Quick Knowledge Check (Mixed Questions)

Multiple Choice:

1. What does HTTPS provide?
   ☐ Encryption
   ☐ Faster internet speed
   ☐ Virus protection

2. What is phishing?
   ☐ A type of malware
   ☐ A social engineering attack
   ☐ A firewall

True / False:

- HTTPS guarantees a website is completely safe (T / F) F
- Strong passwords should include symbols and numbers (T / F) T

Section 4: Reflection

What new cybersecurity concept did you learn?

Vulnerable file types, that include .exe, which is hidden in a normal document.

Which section did you find most difficult?

File safety part due to my lack of knowledge.

How will this activity change your online behaviour?

I will be more careful when executing files.